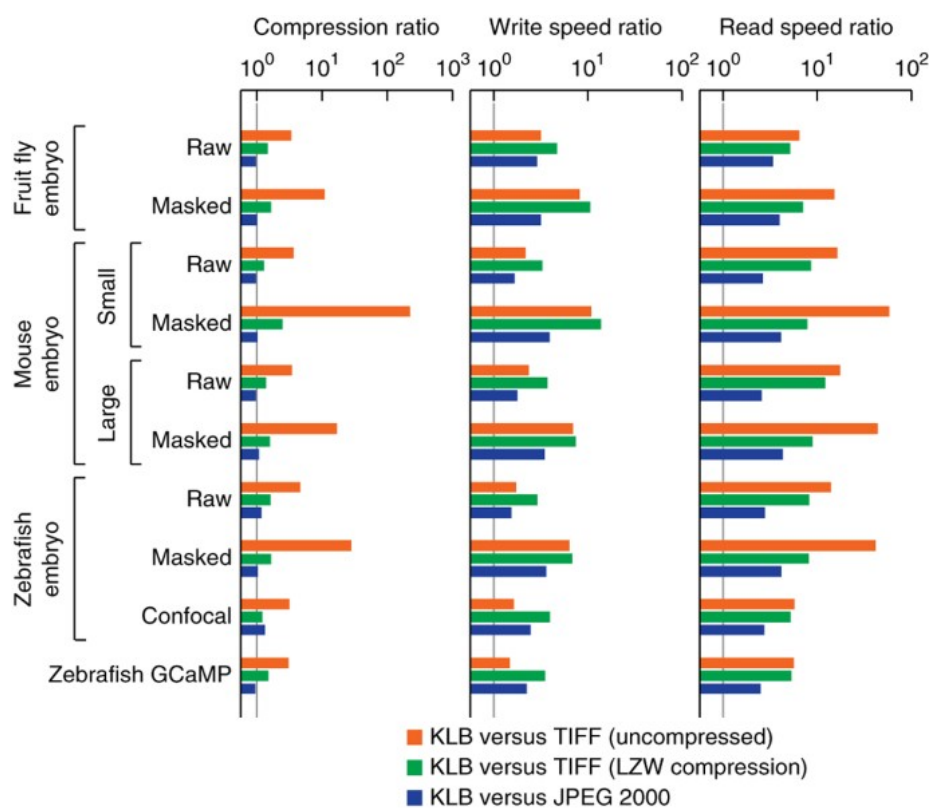
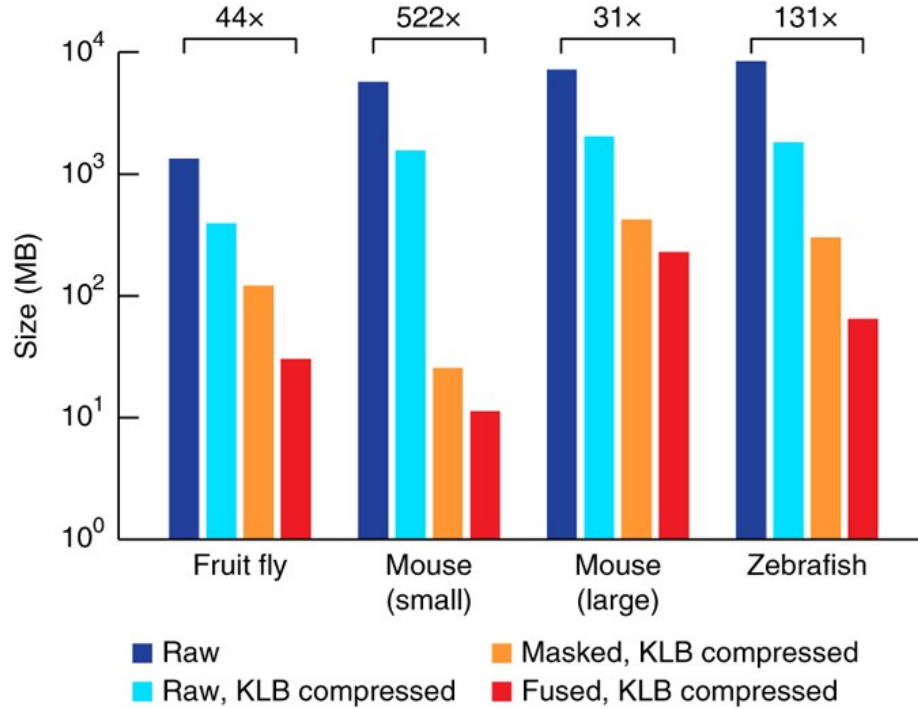


nature [\[\[light_sheet_microscopy/amat_keller_2015/paper\]\]](https://doi.org/10.1038/nature14051)

Key Takeaways

- 5 modules described in [\[\[light_sheet_microscopy/lemon_keller_2015/notes|Lemon et. al., 2015\]\]](https://doi.org/10.1038/nature14051)
- CATMAID + Imaris
- tested on fruit fly, zebrafish, mouse embryo
- Algorithms used with confocal, commercial light sheet
- Requires fast retrieval of ROI, region of the FOV
- Largely describes performance of KLB file format





Comparisons - plane-by-plane is insufficient - can't retrieve arbitrary rois - JPEG is lossless, few 3D compression solutions - HDF5: no parallel write

Multi-view Image Fusion

Initial method is provided as a fiji plugin: - embedded fluor beads surrounding the sample - detects blobs via difference of gaussians - **new** method does not depend on the type of label. - assumes orthogonal views (main limitation) - advantages: - works with large specimen, high magnification - easier prep: don't need to embed in agarose or similar matrix - compensate (partially) for light-refraction (how)

Table 1: Computation Time

Computational Module	Description	Time per	
		Computation Time (s)	Point (s)
clusterPT.m	msCMOS image correction • Background masking • KLB lossless compression	8.36 per time point	8.36
clusterMF.m	Multiview registration • Multiview image fusion	19.49 per 10 time points	1.95

Computational Module	Description	Computation Time Time (s)	Time per Point (s)
localAP.m	• Parameter interpolation	5.09 per experiment	0.04
clusterTF.m	Multiview image fusion	7.88 per time point	7.88
ProcessStack.m	Hierarchical segmentation	2.73 per time point	2.73
TGMM	• Cell tracking • Detection of cell divisions • Filtering of cell lineages	8.29 per time point	8.29

Table 2: Memory Requirements

Computational Module	Module Configuration	Estimated Memory Consumption
clusterPT.m	• sCMOS image correction rotationFlag = 0	$1.2 \times (2n + 2) \times S$
	• Background masking • KLB lossless compression rotationFlag = 0	Up to $1.2 \times (2n + 4) \times S$
clusterMF.m	• Multiview registration Wavelet fusion, 4 views	$13.2 \times S$
	• Multiview image fusion Wavelet fusion, 2 views	Up to $10.8 \times S$
	Other fusion, 4 views	$9.6 \times S$
	Other fusion, 2 views	Up to $8.4 \times S$
clusterTF.m	• Multiview image fusion Wavelet fusion, 4 views	$9.6 \times S$
	Wavelet fusion, 2 views	$7.2 \times S$
	Other fusion, 4 views	$6.0 \times S$
	Other fusion, 2 views	$3.6 \times S$
clusterCS.m	• 3D drift correction	$5.5 \times S$
	• Intensity normalization All settings	

Computational Module	Module Configuration	Estimated Memory Consumption
clusterFR.m	Local background correction All settings	$3.6 \times S$

Image Segmentation

- Most previous methods made for **small organisms**
- Joint segmentation and cell tracking
 - Why do you need this as joint operations?
 - Needs to maintain scalability to big cell counts
- TGMM software scales linearly with n-neurons

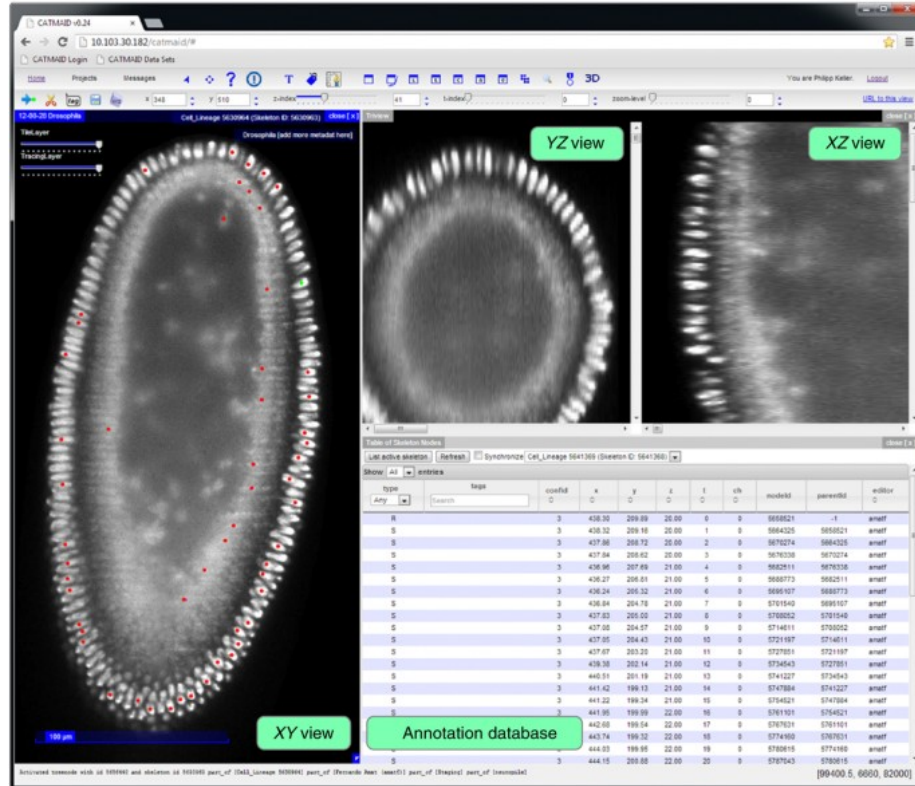
Critical Factor: temporal sampling in the time-lapse data, slightly critical is image quality and cell density

Data Visualization

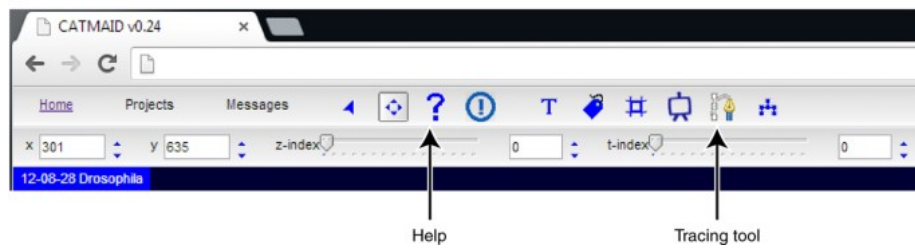
Their software uses a branch of CATMAID - OMERO for data organization - goFigure2 -> VTK library (for rendering) - CATMAIN - similar to goFigure2 - Imaris (Bitplane) - forces large hdf5 which is slow for multiTB datasets - Imaris/goFigure2 allow only a single user to access data

Figure 6: Image annotation and editing of cell-lineage data using CATMAID.

a



b



More software

- Arivis Vision 4D
- Amira
- Vaa3D
- BigDataViewer

CATMAID requires setting up an HTTP sever + PostgreSQL database.

Limitations

- Cannot correct drift larger than the radius of the blob